

Exercises Set: Principal Components Analysis Solutions

DSBA Mathematics Refresher 2026

Abstract

Full solutions. Throughout, variances and covariances use the *sample* convention $\frac{1}{n-1}$ of the notes. Boxes marked *Simplifying the computation* point out the shortcuts that keep everything doable with pen and paper, and say what to do when a number is genuinely not computable by hand.

1 Change of Basis

From $\mathcal{B}'$ to $\mathcal{B}$

Saying $\mathbf{v} = (-1, 3, 2)_{\mathcal{B}'}$ means $\mathbf{v} = -\mathbf{u}_1 + 3\mathbf{u}_2 + 2\mathbf{u}_3$. Since the columns of Q are exactly $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ written in $\mathcal{B}$, this is simply $\mathbf{v}_{\mathcal{B}} = Q\mathbf{v}_{\mathcal{B}'}$:

$$\mathbf{v}_{\mathcal{B}} = - \begin{pmatrix} 0.5 \\ -1 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} + 2 \begin{pmatrix} -0.25 \\ 0.5 \\ 0 \end{pmatrix} = \begin{pmatrix} -0.5 + 6 - 0.5 \\ 1 + 0 + 1 \\ -1 - 3 + 0 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ -4 \end{pmatrix}_{\mathcal{B}}$$

From $\mathcal{B}$ to $\mathcal{B}'$

This is the other direction, $\mathbf{w}_{\mathcal{B}'} = Q^{-1}\mathbf{w}_{\mathcal{B}}$. Rather than inverting Q , solve $Q\mathbf{c} = \mathbf{w}$ directly, which is one small triangular-ish system:

$$\begin{cases} 0.5c_1 + 2c_2 - 0.25c_3 &= -1 \\ -c_1 &+ 0.5c_3 &= 3 \\ c_1 - c_2 &&= 2 \end{cases}$$

The last two equations give $c_1 = \frac{1}{2}c_3 - 3$ and $c_2 = c_1 - 2 = \frac{1}{2}c_3 - 5$. Substituting into the first:

$$\frac{1}{2} \left(\frac{1}{2}c_3 - 3 \right) + 2 \left(\frac{1}{2}c_3 - 5 \right) - \frac{1}{4}c_3 = \frac{1}{4}c_3 + c_3 - \frac{1}{4}c_3 - \frac{3}{2} - 10 = c_3 - \frac{23}{2} = -1$$

so $c_3 = \frac{21}{2}$, then $c_1 = \frac{21}{4} - 3 = \frac{9}{4}$ and $c_2 = \frac{21}{4} - 5 = \frac{1}{4}$:

$$\mathbf{w}_{\mathcal{B}'} = \begin{pmatrix} 2.25 \\ 0.25 \\ 10.5 \end{pmatrix}_{\mathcal{B}'}$$

Sanity check: $2.25\mathbf{u}_1 + 0.25\mathbf{u}_2 + 10.5\mathbf{u}_3 = (1.125 + 0.5 - 2.625, -2.25 + 0 + 5.25, 2.25 - 0.25 + 0)^\top = (-1, 3, 2)^\top$.

Do not invert the 3×3 matrix. Inverting costs a cofactor matrix (nine 2×2 determinants), whereas solving the single system $Q\mathbf{c} = \mathbf{w}$ costs three substitutions. Invert only if you have to change the basis of many vectors. Here the arithmetic stays easy because $\det Q = 1$, which you can check with the rule of Sarrus.

2 Variance and Covariance

Write $\mathcal{S}_1 = \{1.5, 3, 5, 7.5, 8, 9\}$ and $\mathcal{S}_2 = \{0, 2, 4, 6, 8, 10\}$, both of size $n = 6$.

Means

$$\bar{x}_1 = \frac{1.5 + 3 + 5 + 7.5 + 8 + 9}{6} = \frac{34}{6} = \frac{17}{3} \approx 5.667 \quad \bar{x}_2 = \frac{0 + 2 + 4 + 6 + 8 + 10}{6} = 5$$

Deviations

It is worth writing the two lists of deviations once and reusing them for everything else. For $\mathcal{S}_1$, on the common denominator 6:

$$\left(-\frac{25}{6}, -\frac{16}{6}, -\frac{4}{6}, \frac{11}{6}, \frac{14}{6}, \frac{20}{6}\right) \quad (\text{they sum to 0, as they must})$$

For $\mathcal{S}_2$:

$$(-5, -3, -1, 1, 3, 5)$$

Variances

$$s_1^2 = \frac{1}{5} \cdot \frac{25^2 + 16^2 + 4^2 + 11^2 + 14^2 + 20^2}{36} = \frac{1}{5} \cdot \frac{1614}{36} = \frac{269}{30} \approx 8.967$$
$$s_2^2 = \frac{25 + 9 + 1 + 1 + 9 + 25}{5} = \frac{70}{5} = 14$$

Covariance

$$\text{Cov}(\mathcal{S}_1, \mathcal{S}_2) = \frac{1}{5} \cdot \frac{(-25)(-5) + (-16)(-3) + (-4)(-1) + (11)(1) + (14)(3) + (20)(5)}{6} = \frac{1}{5} \cdot \frac{330}{6} = 11$$

Standardized sets

Standardizing means $\hat{x}_i = (x_i - \bar{x})/s$, with $s_1 = \sqrt{269/30} \approx 2.9944$ and $s_2 = \sqrt{14} \approx 3.7417$:

$$\hat{\mathcal{S}}_1 \approx \{-1.391, -0.890, -0.223, 0.612, 0.779, 1.113\}$$

$$\hat{\mathcal{S}}_2 \approx \{-1.336, -0.802, -0.267, 0.267, 0.802, 1.336\}$$

Both now have mean 0 and variance 1.

Covariance of the standardized sets

Do *not* recompute a covariance from the decimals above. Standardizing is an affine map, and covariance is bilinear, so

$$\text{Cov}(\hat{\mathcal{S}}_1, \hat{\mathcal{S}}_2) = \text{Cov}\left(\frac{\mathcal{S}_1 - \bar{x}_1}{s_1}, \frac{\mathcal{S}_2 - \bar{x}_2}{s_2}\right) = \frac{\text{Cov}(\mathcal{S}_1, \mathcal{S}_2)}{s_1 s_2} = \frac{11}{\sqrt{\frac{269}{30}} \sqrt{14}} = \frac{11}{\sqrt{\frac{3766}{30}}} \approx 0.982$$

What do we remark?

The covariance of the standardized data is exactly the **correlation coefficient** ρ of the original data. Three consequences are worth remembering:

- It is dimensionless. The raw covariance 11 is measured in (units of $\mathcal{S}_1$) $\times$ (units of $\mathcal{S}_2$), so its size tells you nothing on its own, whereas 0.982 does.
- It always lies in $[-1, 1]$, by Cauchy-Schwarz. A value of 0.982 says the two sets are very nearly proportional to each other.
- Consequently, **the covariance matrix of standardized data is the correlation matrix of the raw data**, with ones on the diagonal. This is the fact that does all the work in Section 3.

The only step here that a calculator is really needed for is $\sqrt{269/30}$, and the ugliness comes from $\bar{x}_1 = 17/3$ not being an integer. Two ways around it. (i) Never compute $\hat{\mathcal{S}}_1$ and $\hat{\mathcal{S}}_2$ numerically at all: as shown above the last question only needs $\text{Cov}/(s_1 s_2)$, so the square roots appear once, at the very end. (ii) If you want an all-integer version to practise on, replace $\mathcal{S}_1$ by $\{1, 4, 5, 7, 8, 11\}$: the mean is then exactly 6, the deviations are $(-5, -2, -1, 1, 2, 5)$, and $s_1^2 = 12$, $\text{Cov} = 12.8$, $\rho \approx 0.988$. Every qualitative conclusion is unchanged. Also recall that variance and covariance are unchanged by shifting the data, so you may subtract any convenient constant before starting.

3 Principal Component Analysis

(There are 6 points, so $n = 6$ and we divide by $n - 1 = 5$.)

The guess

Before any computation, the left plot shows the six points lying on a perfect straight line, and indeed $x_2 = -0.2x_1$ for every point. Feature x_2 therefore carries *no* information that x_1 does not already carry. The only genuinely new information in the dataset is x_3 : it is the sole coordinate that separates C from D . So we should expect one direction inside the (x_1, x_2) plane to dominate, the x_3 direction to come second, and a third direction carrying strictly nothing. The computation below confirms exactly this.

3.1 Standardization *

Arrange the data as a 6×3 matrix, one row per point. Column means:

$$\bar{x}_1 = \frac{2 + 4 + 12 + 12 + 14 + 16}{6} = \frac{60}{6} = 10, \quad \bar{x}_2 = \frac{-12}{6} = -2, \quad \bar{x}_3 = \frac{0}{6} = 0$$

Centered data (subtract the means):

$$X_c = \begin{pmatrix} -8 & 1.6 & 0.1 \\ -6 & 1.2 & -0.1 \\ 2 & -0.4 & -0.5 \\ 2 & -0.4 & 0.5 \\ 4 & -0.8 & -0.1 \\ 6 & -1.2 & 0.1 \end{pmatrix}$$

Note immediately that the second column is -0.2 times the first. Sums of squared deviations, then variances:

$$\sum (x_1 - \bar{x}_1)^2 = 64 + 36 + 4 + 4 + 16 + 36 = 160 \quad \Rightarrow \quad s_1^2 = \frac{160}{5} = 32, \quad s_1 = 4\sqrt{2} \approx 5.657$$

$$\sum (x_2 - \bar{x}_2)^2 = (0.2)^2 \times 160 = 6.4 \quad \Rightarrow \quad s_2^2 = \frac{6.4}{5} = 1.28, \quad s_2 = 0.2 \times 4\sqrt{2} \approx 1.131$$

$$\sum (x_3 - \bar{x}_3)^2 = 0.01 + 0.01 + 0.25 + 0.25 + 0.01 + 0.01 = 0.54 \quad \Rightarrow \quad s_3^2 = \frac{0.54}{5} = 0.108, \quad s_3 = \frac{3\sqrt{30}}{50} \approx 0.3286$$

Dividing each column of X_c by its standard deviation:

$$\hat{S} = Z = \begin{pmatrix} -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}}{4} & \frac{\sqrt{30}}{18} \\ -\frac{3\sqrt{2}}{4} & \frac{3\sqrt{2}}{4} & -\frac{\sqrt{30}}{18} \\ \frac{\sqrt{2}}{4} & -\frac{\sqrt{2}}{4} & -\frac{5\sqrt{30}}{18} \\ \frac{\sqrt{2}}{4} & -\frac{\sqrt{2}}{4} & \frac{5\sqrt{30}}{18} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & -\frac{\sqrt{30}}{18} \\ \frac{3\sqrt{2}}{4} & -\frac{3\sqrt{2}}{4} & \frac{\sqrt{30}}{18} \end{pmatrix} \approx \begin{pmatrix} -1.414 & 1.414 & 0.304 \\ -1.061 & 1.061 & -0.304 \\ 0.354 & -0.354 & -1.521 \\ 0.354 & -0.354 & 1.521 \\ 0.707 & -0.707 & -0.304 \\ 1.061 & -1.061 & 0.304 \end{pmatrix}$$

The second column is now exactly the opposite of the first, which is the standardized form of the relation $x_2 = -0.2x_1$.

*These irrational entries are unpleasant, and the good news is that **you never need them**. Sections 3.2 to 3.5 can all be done from the centered data X_c and the three standard deviations, using $\text{Cov}(\hat{x}_i, \hat{x}_j) = \text{Cov}(x_i, x_j)/(s_i s_j)$ from Section 2. Keep $\hat{S}$ in the symbolic form above, or skip it entirely, and never convert to decimals until the final plot.*

3.2 Covariance matrix *

First the covariances of the *raw* features, using the centered columns:

$$\text{Cov}(x_1, x_2) = \frac{1}{5} \sum (x_1 - \bar{x}_1)(x_2 - \bar{x}_2) = \frac{1}{5} \times (-0.2) \times 160 = -6.4$$

$$\begin{aligned}\text{Cov}(x_1, x_3) &= \frac{(-8)(0.1) + (-6)(-0.1) + (2)(-0.5) + (2)(0.5) + (4)(-0.1) + (6)(0.1)}{5} \\ \text{Cov}(x_1, x_3) &= \frac{-0.8 + 0.6 - 1 + 1 - 0.4 + 0.6}{5} = 0 \\ \text{Cov}(x_2, x_3) &= -0.2 \times \text{Cov}(x_1, x_3) = 0\end{aligned}$$

so that

$$\Sigma_{\text{raw}} = \begin{pmatrix} 32 & -6.4 & 0 \\ -6.4 & 1.28 & 0 \\ 0 & 0 & 0.108 \end{pmatrix}$$

Now divide entry (i, j) by $s_i s_j$ to get the covariance matrix of the standardized data:

$$\text{Cov}(\hat{x}_1, \hat{x}_2) = \frac{-6.4}{4\sqrt{2} \times 0.8\sqrt{2}} = \frac{-6.4}{6.4} = -1, \quad \text{Cov}(\hat{x}_1, \hat{x}_3) = \text{Cov}(\hat{x}_2, \hat{x}_3) = 0$$

and the diagonal is 1 by construction. Hence the matrix we work with is the correlation matrix:

$$C = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The value -1 is the algebraic statement of what the left plot showed: x_1 and x_2 are perfectly anti-correlated, and C is singular.

3.3 Eigenvalues of the covariance matrix *

C is block-diagonal, so the characteristic polynomial factors immediately:

$$\begin{aligned}\det(C - \lambda I) &= \det \begin{pmatrix} 1 - \lambda & -1 \\ -1 & 1 - \lambda \end{pmatrix} \times (1 - \lambda) = [(1 - \lambda)^2 - 1] (1 - \lambda) = (\lambda^2 - 2\lambda)(1 - \lambda) \\ \det(C - \lambda I) &= -\lambda(\lambda - 2)(\lambda - 1) = 0 \implies \lambda_1 = 2, \quad \lambda_2 = 1, \quad \lambda_3 = 0\end{aligned}$$

Check: the eigenvalues sum to $3 = \text{tr}(C)$, and $\lambda_3 = 0$ because C is singular. Explained variance:

$$\frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3} = \frac{2}{3} \approx 66.7\%, \quad \frac{\lambda_2}{3} = \frac{1}{3} \approx 33.3\%, \quad \frac{\lambda_3}{3} = 0\%$$

Ordered by decreasing importance: the first component carries two thirds of the variability, the second one third, and the third carries nothing at all. The dataset, although given in $\mathbb{R}^3$, really lives in a plane.

3.4 Feature vectors (the “principal components”) *

$\lambda_1 = 2$

$$\begin{aligned}(C - 2I)\mathbf{u} &= \begin{pmatrix} -1 & -1 & 0 \\ -1 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \mathbf{0} \iff \begin{cases} u_1 + u_2 = 0 \\ u_3 = 0 \end{cases} \\ \mathbf{u}_1 &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}\end{aligned}$$

$\lambda_2 = 1$

$$(C - I)\mathbf{u} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \mathbf{u} = \mathbf{0} \iff u_1 = u_2 = 0 \implies \mathbf{u}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\lambda_3 = 0$$

$$C\mathbf{u} = \mathbf{0} \iff \begin{cases} u_1 - u_2 = 0 \\ u_3 = 0 \end{cases} \implies \mathbf{u}_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

The three are pairwise orthogonal, as they must be for a symmetric matrix with distinct eigenvalues. Their interpretation is pleasant: $\mathbf{u}_1$ is “ $\hat{x}_1$ goes up while $\hat{x}_2$ goes down”, the direction along which the six points are spread out; $\mathbf{u}_2$ is the x_3 axis, untouched because x_3 is uncorrelated with the other two; and $\mathbf{u}_3$ is “ $\hat{x}_1$ and $\hat{x}_2$ go up together”, which never happens in this dataset, hence $\lambda_3 = 0$.

3.5 Recasting data on principal components axes *

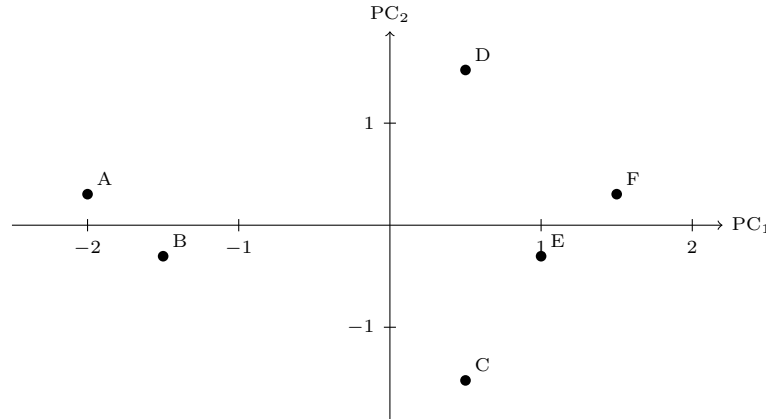
Projecting means multiplying the standardized data by each feature vector. Because the second column of Z is the opposite of the first, we get for the first component

$$Z\mathbf{u}_1 = \frac{\hat{x}_1 - \hat{x}_2}{\sqrt{2}} = \frac{2\hat{x}_1}{\sqrt{2}} = \sqrt{2} \hat{x}_1 = \sqrt{2} \times \frac{x_1 - 10}{4\sqrt{2}} = \frac{x_1 - 10}{4}$$

which is a whole number of quarters, no square roots left. The second component is just $Z\mathbf{u}_2 = \hat{x}_3$, and the third is $Z\mathbf{u}_3 = (\hat{x}_1 + \hat{x}_2)/\sqrt{2} = 0$ for every point.

	PC ₁	PC ₂	PC ₃
A	-2.0	0.304	0
B	-1.5	-0.304	0
C	0.5	-1.521	0
D	0.5	1.521	0
E	1.0	-0.304	0
F	1.5	0.304	0

Check: the variance of the first column is $\frac{4+2.25+0.25+0.25+1+2.25}{5} = \frac{10}{5} = 2 = \lambda_1$, and the variance of the second is $\frac{5}{5} = 1 = \lambda_2$, as promised. The third column is identically zero, so **nothing at all is lost by dropping the third component**: the two-dimensional plot below is a faithful, complete representation of the original three-dimensional data.



The six points in the plane of the first two principal components. C and D, which were indistinguishable on the original (x_1, x_2) plot, are now the two most widely separated points along PC_2 .

3.6 Importance of standardization

Redo the same steps starting from Σ_{raw} instead of C .

Eigenvalues

The matrix is again block-diagonal, and the top-left block is singular, since $32 \times 1.28 - (-6.4)^2 = 40.96 - 40.96 = 0$. A 2×2 matrix with determinant 0 has eigenvalues 0 and its trace, so the block gives 0 and $32 + 1.28 = 33.28$, and the last diagonal entry gives 0.108:

$$\lambda_1 = 33.28, \quad \lambda_2 = 0.108, \quad \lambda_3 = 0, \quad \text{total } 33.388$$

$$\frac{33.28}{33.388} \approx 99.68\%, \quad \frac{0.108}{33.388} \approx 0.32\%, \quad 0\%$$

Feature vectors

For $\lambda_1 = 33.28$, the first row of $\Sigma_{\text{raw}} - \lambda_1 I$ reads $-1.28 u_1 - 6.4 u_2 = 0$, that is $u_1 = -5u_2$:

$$\mathbf{u}_1 = \frac{1}{\sqrt{26}} \begin{pmatrix} 5 \\ -1 \\ 0 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{u}_3 = \frac{1}{\sqrt{26}} \begin{pmatrix} 1 \\ 5 \\ 0 \end{pmatrix}$$

Projections

Projecting the *centered but unscaled* data:

$$X_c \mathbf{u}_1 = \frac{5(x_1 - 10) - (x_2 + 2)}{\sqrt{26}} = \frac{5.2(x_1 - 10)}{\sqrt{26}} \approx 1.020(x_1 - 10)$$

	PC ₁	PC ₂
A	-8.16	0.1
B	-6.12	-0.1
C	2.04	-0.5
D	2.04	0.5
E	4.08	-0.1
F	6.12	0.1

Conclusion

The two analyses give the same *directions* but wildly different *importances*:

- Without standardization, the first component absorbs 99.68% of the variance and is essentially the x_1 axis. The reason has nothing to do with structure in the data: x_1 simply happens to be recorded in units that make its numbers large. Multiply x_1 by 1000 (say, measure it in grams instead of kilograms) and its share would rise to 99.9999%.
- In that unstandardized view, x_3 looks like negligible noise at 0.32%, and a practitioner reading a scree plot would discard it without hesitation. Yet x_3 is the only feature carrying independent information, the only thing distinguishing C from D . Discarding it destroys the dataset, while discarding x_2 , which appears important, costs nothing.
- After standardization every feature starts on equal footing, and the answer becomes honest: two thirds for the x_1/x_2 direction, one third for x_3 , zero for the redundant direction.

The moral: PCA maximizes variance, and variance depends on units. Standardize whenever the features are not measured in comparable units, which in practice is almost always.

Three tricks make the whole of Section 3 a pen-and-paper exercise. (i) Spot the exact linear relation $x_2 = -0.2x_1$ at the start. It tells you before any computation that the correlation is -1 , that the covariance matrix is singular, and that one eigenvalue is 0. (ii) Use $\text{Cov}(\hat{x}_i, \hat{x}_j) = \text{Cov}(x_i, x_j)/(s_i s_j)$ so that the irrational standard deviations cancel and the correlation matrix comes out with entries 0 and ± 1 . (iii) Exploit the block structure: a matrix of the form $\begin{pmatrix} a & b & 0 \\ b & c & 0 \\ 0 & 0 & d \end{pmatrix}$ has d as an eigenvalue for free, and the remaining 2×2 block has characteristic polynomial $\lambda^2 - (a+c)\lambda + (ac-b^2)$, so no cubic ever needs to be solved. If you want to avoid decimals entirely, rescale x_3 by a factor 10 before starting. Standardization removes any scale factor, so the standardized answers of Sections 3.1 to 3.5 are strictly unchanged.

4 Principal Component Analysis with Python

See `NotebookPCA_solutions.ipynb` in this folder (a PDF export, `NotebookPCA_solutions.pdf`, is provided as well).